

LONG-TERM FOLLOW-UP AFTER UVEAL MELANOMA CHARGED PARTICLE THERAPY*

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ABSTRACT

Purpose: To examine the results of helium ion irradiation in 218 uveal melanoma patients treated more than 10 years ago.

Methods: Retrospective review of 218 eyes treated with helium ion radiation for uveal melanoma between 1978 and 1984. Several parametric and non-parametric statistical analysis techniques were used.

Results: In 218 eyes treated with helium ion radiation for uveal melanoma, the mean dimension for largest basal diameter was 11.9 mm (range 5 mm to 24 mm). The mean tumor thickness was 6.7 mm (range 1.3 mm to 14.2 mm). Following helium ion radiation 208 (95.4%) of 218 eyes had local tumor control. At 10 years after radiation 46 (22.4%) of 218 eyes were enucleated; the majority (37 of 46) of enucleations were due to anterior ocular segment complications. At 10 years after radiation 102 (46.8%) of the 218 patients were dead; half had non-tumor related deaths and 51 died from metastatic melanoma. Best corrected visual acuity after radiation was $\geq 20/40$ in 21 of 93 eyes of patients that were alive and retained their eyes 10 or more years after treatment. In patients with tumors that were less than 6 mm in height and more than 3 mm away from the nerve or the fovea, 13 of 18 (72%) retained $\geq 20/40$. In contrast, only 11% of the patients with either thicker tumors or those close to the nerve or fovea retained that level of acuity. The actuarial enucleation rate at 5 years was 17.2% (2.7% S.E.) and at 10 years this was 22.4% (3.1% S.E.). The recurrence tumor control rate at both 5 and 10 years was 5.3% (S.E. 1.7%).

Conclusions: Helium ion radiation of uveal melanoma is associated with good local tumor control and reasonable retention of the treated eye 10 years after treatment. In tumors that are less than 6 mm in thickness and greater than 3 mm from the optic nerve and fovea, many retain excellent

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vision. Approximately one-half of the deaths 10 years after treatment were due to non-tumor-related causes.

INTRODUCTION

There is a paucity of long-term follow-up data on irradiated uveal melanoma patients.¹ Many reports have only 1 to 2 years of post-treatment data, and only a few have actuarial analyses of longer time periods.²⁻¹⁰ Several of these analyses, including our own, have demonstrated that both enucleated and irradiated uveal melanoma patients share most prognostic parameters, with the caveat that in irradiated patients, histologic data are usually not available.¹¹ In the irradiated group, an additional prognostic parameter is local tumor control; intraocular tumor failure after radiation is associated with increased tumor-related mortality.^{8,12-15}

Unfortunately, unlike some ophthalmic malignancies, such as rhabdomyosarcoma, in which almost all tumor-related deaths occur within 2 years, uveal melanoma metastases have been reported to occur over 40 years after ocular therapy.^{1,16} We have previously reported our 5-year post-treatment data in patients treated with helium ion radiation.¹⁷ In that report and ones from other centers, the overall 5-year tumor-related mortality after treatment was less than 20%; in cases with melanomas less than 16-mm diameter and less than 8-mm thickness, it was under 10%.¹⁷ Some posttreatment complications, such as neovascular glaucoma, are always manifest within 5 years after irradiation, while others, such as cataract and melanoma-associated mortality, continue to develop.¹⁸ In this study, we analyzed the 10-year posttreatment results after charged particle radiation.

PATIENTS AND METHODS

All 221 uveal melanoma patients treated with helium ion irradiation prior to January 1985 were eligible for the study. The first treatment was completed in February 1978 and the last treatment in December 1984. Three patients were excluded from analyses; 1 patient had bilateral presentation, and 2 patients had simultaneously detected metastatic disease. The remaining 218 patients were included in all analyses. Several different phase I-III radiation protocols were used to treat the patients. Therapy was delivered in either 4 or 5 fractions to a homogeneous tumor dose of 50, 60, 70, or 80 GyE.^{17,19,20}

All patients were examined and treated at the University of California, San Francisco, School of Medicine after multimodality, multiobserver evaluation.²¹ All patients had standardized follow-up of both intraocular status (clinical examination, eye photography, and ultrasonography) and

medical investigations to detect metastatic disease (general physical examination, serum lactic dehydrogenase and glutamyl transpeptidase measurements, and chest x-ray) at routine intervals. If any of the later tests were abnormal, further studies were obtained. Generally, these tests were performed 4 times in the first year after therapy, 3 times in the second year after radiation, twice a year for 2 additional years, and then yearly.

Details of the deaths (the apparent cause of death and method of detection) were obtained following the initial notification of a patient death by telephone conversations with family members, and from regional cancer registries, primary physicians, or the patient's medical oncologist and death certificates. Melanoma-related mortality was coded as the cause of death in any patient who died with a known history of metastatic melanoma. In patients who died of a different neoplastic process, the hospital records were obtained. If the cause of death was equivocal, histologic materials were reviewed.

Death certificates were obtained with the assistance of the individual state vital statistics departments. Death certificates were obtained for 92 of the 102 (90.2%) deaths. Missing death certificates were principally attributed to foreign citizenship or recent deaths. Five of the 10 cases without death certificates were classified as melanoma metastatic deaths on the basis of known metastatic disease and conversations with family and medical oncologist.

Current follow-up was available in all cases. The 218 eligible cases had 2,046 person-years of follow-up; average follow-up length was 9.4 years. The 116 patients alive at last status had mean follow-up of 12.5 years and median follow-up of 12.4 years.

Statistical methods were based on counting process approaches to censored survival data.²² Kaplan-Meier techniques were used to estimate event rates and quantiles.²³ Cox proportional hazards models were used to examine the effects of covariates on the time to events.²⁴ Cumulative incidence estimates, which estimate the proportion of patients who are expected to experience an event in the presence of competing risks (ie, non-melanoma mortality) at any time point were used.²⁵ The cumulative incidence estimates the proportion of patients who are expected to experience an event. These differ from the Kaplan-Meier analysis, which estimates the proportion of patients expected to experience an event if they are never censored from other causes. Expected mortality was estimated from United States age-, sex-, and race-specific life table estimates of annual mortality using the person-years method.^{26,27} Mortality rates for 1989 (the median time point) were used for all follow-up years for computational convenience. Published annual mortality probabilities were transformed to hazard rates by assuming constant annual hazard rates. In analyses of mortality and enucleation, observations were censored at the

last patient contact. In analyses of cataract and neovascular glaucoma, observations were censored at their last ophthalmologic visit. In the analysis of tumor control, observations were censored at the last ophthalmologic visit in which the tumor was viewable or quantitative ultrasound was performed. In analyses of visual function, observations were censored at their last assessment of visual acuity.

RESULTS

Patient and tumor specific characteristics were prospectively collected on all patients and summarized in Table I. Patient statistics included sex (47% female), race (97.7% Caucasian), age (mean, 58.0 years; SD, 14.1 years), eye (54.6% right), and presence of symptoms before treatment (84.9% symptomatic). Tumor statistics included largest tumor diameter (mean, 11.9 mm; SD, 3.7 mm; range, 5 to 24 mm), opposing diameter (mean, 9.6 mm; SD, 2.8 mm; range, 3.5 mm to 20 mm), ultrasound tumor thickness (mean, 6.7 mm; SD, 2.7 mm; range, 1.3 mm to 14.2 mm), minimum tumor distance to optic nerve (mean, 5.8 mm; SD, 4.0 mm), distance to fovea (mean, 5.4 mm; SD, 4.3 mm), location vis a vie the equator (43.1% posterior), ciliary body involvement (24.3% involved), and documented pretreatment growth (29.8%).

Ten (4.6%) of the tumors had neoplastic tumor enlargement (definition of tumor recurrence) following therapy. Local tumor recurrence was almost always demonstrable with at least 2, and often all 3, methods of size measurement (clinical examination, photographs, and ultrasonography) using the same instruments and methods of measurement during the entire study.²¹ The 5- and 10-year Kaplan-Meier estimates of tumor recurrence were 5.3% (SE = 1.7%). Three cases were retreated with helium ions, 6 eyes were enucleated, and 1 case was treated by laser for a flat horizontal recurrence. Seven recurrences were within 1 year of treatment. The other 3 recurrences occurred 1.4, 3.1, and 4.9 years after treatment. No local recurrences have occurred in the over 1,000 person-years of ophthalmologic follow-up at least 5 years after treatment.

Eye retention estimates and confidence intervals for enucleation are given in Fig 1. The 5- and 10-year Kaplan-Meier estimates were 17.2% (SE = 2.7%) and 22.4% (SE = 3.1%). Overall, 46 eyes were enucleated. Thirty-five (76.1%) were due to anterior segment complications related to neovascular glaucoma or corneal disease. Six eyes were enucleated following local recurrence; 5 eyes were enucleated for pain or patient selection. The hazard for enucleation is largest early after treatment and decreases with time. Over half (24 of 46) of the enucleations were within 2 years of treatment. Three enucleations occurred after 10 years.

One hundred two (46.8%) patients died. Kaplan-Meier survival esti-

TABLE I. PATIENT AND TUMOR SUMMARY

VARIABLE	GROUP	No.	%
Gender	Male	116	53.2
	Female	102	46.8
Age at tumor treatment (yr)	<40	30	13.8
	40-50	31	14.2
	50-60	44	20.2
	60-70	68	31.2
	>70	45	20.6
Tumor -related symptoms	Present	185	84.9
	Absent	33	15.1
Documented tumor growth	Yes	65	29.8
	No	153	70.2
Year of treatment	1978	4	1.8
	1979	10	4.6
	1980	19	8.7
	1981	21	9.6
	1982	38	17.4
	1983	54	24.8
	1984	37	17.0
Tumor (GyE)	50	23	10.6
	60	61	28.0
	70	66	30.3
	80	68	31.2
Largest tumor diameter (mm)	≤8	33	15.1
	8 - 10	38	17.4
	10 - 12	68	31.2
	12 - 14	34	15.5
	14 - 16	15	6.9
	> 16	30	13.8
Ultrasound tumor height (mm)	≤4	41	18.8
	4 - 6	59	27.1
	6 - 8	45	20.6
	8 - 10	45	20.6
	> 10	28	12.8
Minimum distance to nerve or fovea (mm)	<3	94	43.1
	3 - 5	36	16.5
	5 - 7	30	13.8
	≥7	58	26.6
Posterior tumor border	Posterior equator	184	84.4
	Anterior equator	32	14.7
	Ciliary body	2	0.9
Anterior tumor border	Posterior equator	94	43.1
	Anterior equator	71	32.6
	Ciliary body	53	24.3

Enucleation

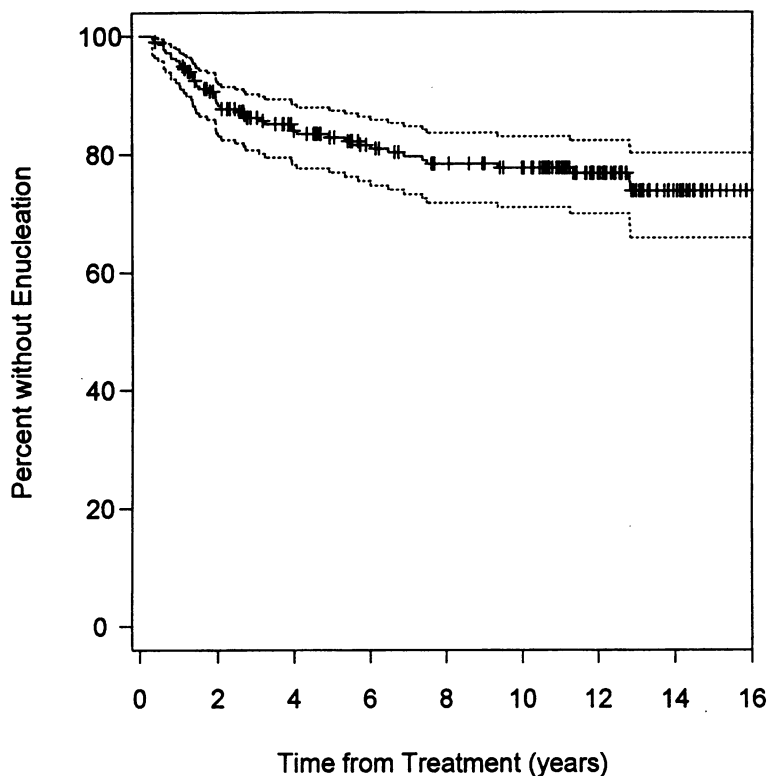


FIGURE 1

Kaplan-Meier estimate of percent of eyes undergoing enucleation after helium ion radiation for uveal melanoma. Dashed lines represent 95% confidence intervals. Hash marks correspond to censored observations.

mates and 95% confidence intervals are given in Fig 2. The 5-, 10-, and 15-year mortality rates were 27.5% (SE = 3.0%), 39.0% (SE = 3.3%), and 54.4% (SE = 5.7%). Half of the deaths, 51 (23.4%), were classified as melanoma-related and the other half as non-melanoma-related. Of the non-melanoma deaths, 24 were classified as related to other cancers, 16 to diseases of the circulatory system, and 11 to accidents, respiratory illness, infection, kidney disease, or cerebral causes. The expected number of deaths in a matched cohort was 48.45. The standardized mortality ratio for all deaths was 2.11. The standardized mortality ratio for all non-

All Cause Mortality

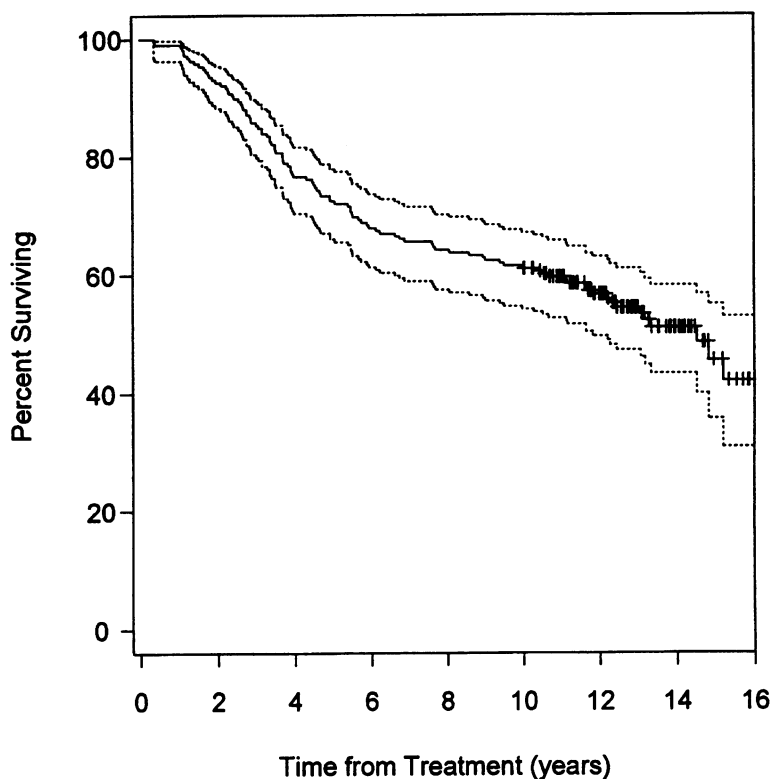


FIGURE 2

Kaplan-Meier estimate of percent of patient deaths from any cause. Dashed lines represent 95% confidence intervals. Hash marks correspond to censored observations.

melanoma patient deaths was 1.05 and was not statistically different than 1 ($P > 0.71$).

Fifty-six patients developed detectable melanoma metastases. Fifty-one of the 56 have died of the disease. The liver was involved in 49 of the 56 cases. The 5- and 10-year melanoma mortality Kaplan-Meier estimates were 18.6% and 23.6%. The cumulative incidence estimates were 17.3% and 21.4%. Results from multivariate Cox proportional hazards model (with non-tumor-related deaths censored at the time of death) demonstrated that largest tumor diameter was the dominant prognostic factor.

Older patient age and anterior tumor location were additional multivariate prognostic factors. A univariate plot by largest tumor diameter is shown in Fig 3. For tumors less than or equal to 12 mm in diameter, the 10-year Kaplan-Meier estimate for metastatic melanoma onset was 13.0% (SE= 3.0%). For tumors larger than 12 mm in diameter, the 10-year Kaplan-Meier estimate for metastatic melanoma onset was 45.1% (SE = 6.0%). These estimates ignore the deaths from other causes. They estimate the probability of detectable metastatic melanoma onset before 10 years while not dying of other causes. The cumulative incidence estimates representing the probability of metastatic melanoma onset within 10 years and accounting for deaths from other causes were 10.7% and 39.7%, respectively.

Significant cataracts were defined as greater than or equal to a clini-

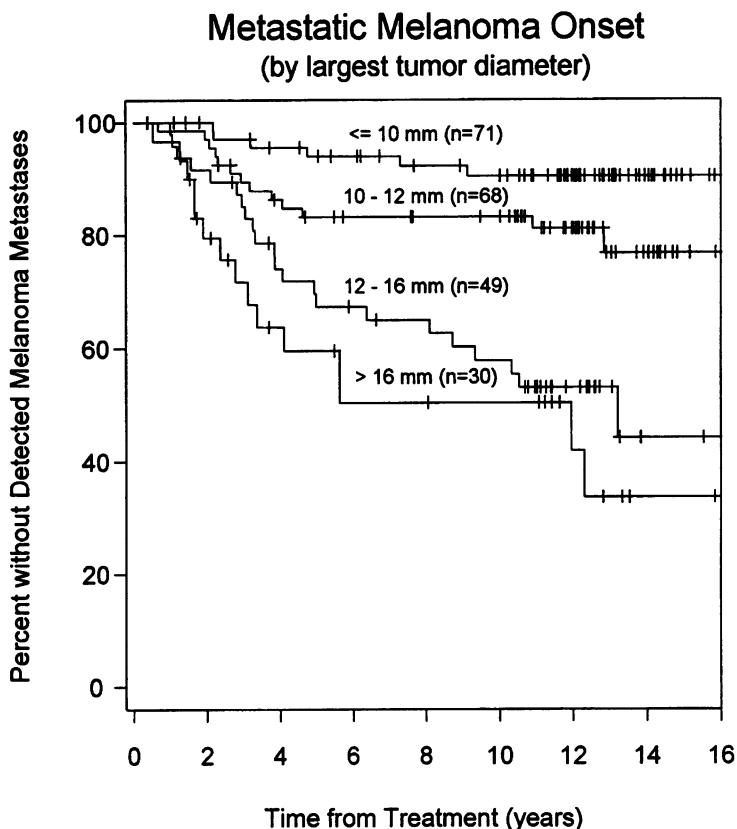


FIGURE 3

Kaplan-Meier univariate estimates of percent of patients with detected uveal melanoma metastases stratified by largest tumor diameter. Hash marks correspond to censored observations.

cally 3+ cataract (on a scale of 0 to 4+) that reduced the visual acuity to 20/100 or worse or resulted in cataract extraction. Nine patients with pretreatment significant cataract or pretreatment intraocular lenses were excluded from the analysis. The most predictive factor for the time to significant cataract was the percent of lens included in the port. Older patients and patients with tumors involving the ciliary body had additional significant increased relative risk. Radiation tumor dose was not a significant multivariate factor. Kaplan-Meier estimates by quartiles of percentage of lens in the treatment port are given in Fig 4. With the exception of 4 patients who were younger than 50 years at treatment and who developed lesser cataracts, significant cataracts developed within 7 years in all patients $\geq 50\%$ of lens in the treatment port. Fifty-one patients under-

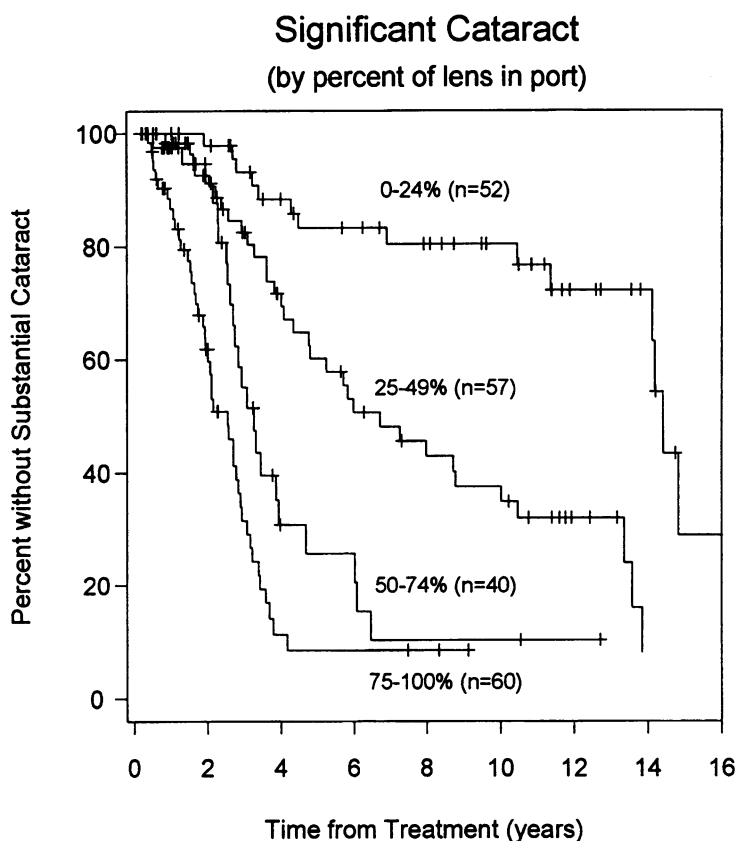


FIGURE 4

Kaplan-Meier univariate estimates of percent of patients with significant cataract stratified by percent of lens in port. Hash marks correspond to censored observations

went cataract extraction. Thirty-two of the 93 patients alive and without enucleation have undergone cataract extraction. The other 61 patients had no asymmetric cataract present (8), cataract of grade 2 or less (26), or substantial cataracts present (27).

Neovascular glaucoma developed in 76 (34.9%) of the 218 patients. Neovascular glaucoma was a severe complication; 37 of the 76 patients who developed it had later enucleations, which accounted for 80% of the 46 enucleations. The most predictive factor for the time to neovascular glaucoma was the percent of lens included in the port. Patients with thicker tumors and patients with anterior tumor locations had additional significant increased relative risk. Treatment dose was not a significant multivariate factor. Kaplan-Meier estimates by quartiles of percentage of lens in the treatment port are given in Fig 5.

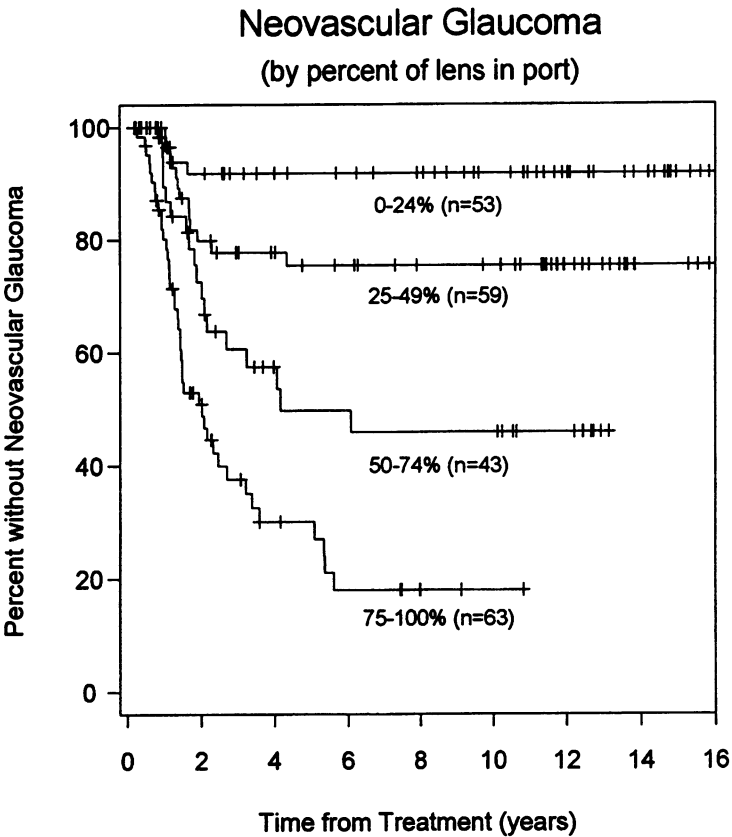


FIGURE 5
Kaplan-Meier univariate estimates of percent of patients with neovascular glaucoma stratified by percent of lens in port. Hash marks correspond to censored observations.

The timing of visual loss is displayed in Fig 6. The slopes of the cumulative hazard lines give the rate of visual loss. The rate of visual loss was most rapid immediately following treatment, dropping between 0.25 to 3 years and further dropping after 3 years. The 3 most predictive factors for the rate of visual loss were tumor size (predominantly tumor thickness), tumor location (predominantly the distance to the optic nerve or fovea) and pretreatment visual acuity. For example, the relative risk for a 3 Bailey-Lovie line loss (ie, a halving of the Snellen visual acuity) was 1.23 (95% CI = 1.15-1.32, $P < 0.0001$) per millimeter of tumor thickness and 1.10 (95% CI = 1.05-1.15, $P < 0.0001$) per millimeter of proximity to the nerve or fovea.

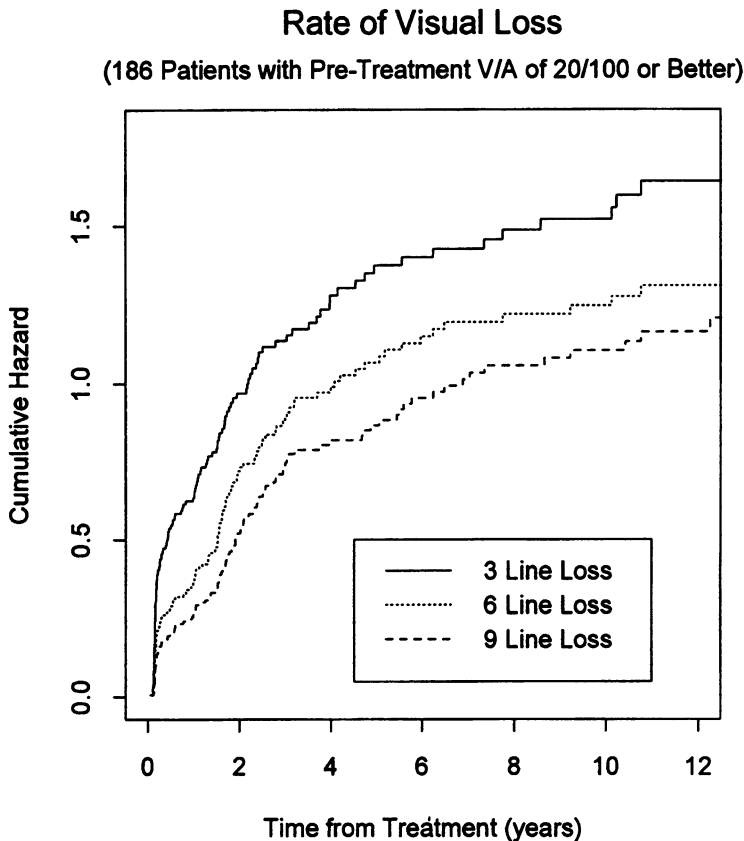


FIGURE 6

Nelson-Aalen cumulative hazard estimates of time to 3, 6, and 9-line loss on Bailey-Lovie chart. Three-line loss corresponds to a doubling of minimal angle of resolution. Slopes of cumulative hazard lines reflect rate (ie., hazard rate) of visual loss.

The primary causes for permanent visual loss (of at least 3 lines) were classified into 5 categories (Table II): (1) posterior pole retinopathy or cystoid macular edema, (2) exudative retinal detachment, (3) anterior segment complications such as cataract, glaucoma, or corneal abnormalities, (4) vitreous hemorrhage, or (5) local tumor recurrence with accompanying exudates or fluid. In almost all cases, it was readily apparent which was the most important cause of visual loss. In a few cases, more than one etiology for vision loss was eventually present, but the first cause of visual decrease (posterior pole retinopathy) was chosen even though the patients developed cataract and neovascular glaucoma. The time to visual loss was stratified into 3 time periods: less than 1 year after treatment, between 1 and 4 years after treatment, and over 4 years after treatment. Within 1 year, 61 patients had a visual loss (Table II). The majority (62%) were caused by exudative retinal detachments. Between 1 and 4 years after treatment, 56 patients had visual loss onset, which was usually related to anterior segment complications (45%) or radiation retinopathy/cystoid macular edema (43%). After 4 years, almost all of the 15 cases of visual loss onset were related to radiation retinopathy.

TABLE II. ONSET AND CAUSE OF VISUAL LOSS AFTER UVEAL MELANOMA TREATMENT WITH HELIUM ION RADIATION

ASSIGNED CAUSE	TIME OF VISUAL LOSS					
	0-1 Yr		1-4 Yr		<4 Yr	
	No.	%	No.	%	No.	%
Radiation retinopathy/ cystoid macular edema	8	13.1	24	42.9	13	86.7
Anterior segment complications	6	9.8	25	44.6	1	6.7
Retinal detachment	38	62.3	2	3.6	0	0
Vitreous hemorrhage	4	8.2	5	8.9	1	6.7
Local recurrence with exudates/fluid	5	8.2	0	0	0	0
Total	61	100	56	100	15	100

At the time of analysis, 102 patients had died and 23 additional patients had undergone enucleation. Of the 93 patients remaining 10 years or more after treatment, 21 had best-corrected visual acuities of 20/40 (23%) or better, 10 (11%) had visual acuities between 20/50 and 20/200, and 62 (66%) had visual acuities of 20/400 or worse at their last ophthalmic examination. The mean and median follow-up for the 93 patients was 11.2 years. While 13 (72%) of 18 patients with tumor less than 6 mm in height and more than 3 mm from the nerve or fovea retained 20/40 or better vision at last follow-up, 8 (11%) of the 75 patients with tumors greater than 6 mm in height or less than 3 mm from the nerve or fovea retained 20/40 vision or better at last follow-up.

DISCUSSION

In this and previous reports from our institution and others, excellent local tumor control has been documented with charged particle radiation of uveal melanomas.^{10,17,28} Local tumor control (defined as either tumor shrinkage or stability after radiation therapy) was achieved in approximately 95% of cases from both institutions. Local failures have been due to an unsuspected diffuse melanoma, radioresistant tumors, or technical errors that resulted in incomplete treatment of the tumor.^{17,28} Retrospective analysis has demonstrated a better local control rate with charged particles as compared with radioactive plaques, and in a prospective dynamically balanced study we also observed a statistically significant difference.^{17,20,28,29} Local control with particles is much better than with brachytherapy, especially with thinner tumors near the optic disc.²⁰

Several investigators have raised concerns about possible late regrowth, since most irradiated uveal melanomas shrink but are not entirely dissipated after therapy. In prostate cancer radiation, some late recurrences have been noted, and the question had been raised whether particle radiation produced only temporary tumor control.^{30,31} In this data set with patients' follow-up over 10 years after treatment, no local recurrences developed more than 5 years after treatment. It is most likely that local radiation tumor control is permanent in these long-term cases with over 1,000 patient-years of follow-up.

We and others have noted that most prognostic factors relating to melanoma-associated mortality are similar in enucleated and irradiated cases.¹ In irradiated cases, more malignant fine-needle biopsy cytology data and absence of local control also have a negative impact on survival.^{11,12,14,15,32} As others have noted, maximum tumor diameter is the dominant prognostic risk factor. In tumors 12 mm or less in diameter, the 10-year metastatic rate was 13%, but this rose to 45% for larger melanomas. In this study, with 100% follow-up, it is important to emphasize that half

of the deaths observed were from non-melanoma-related causes. Our patients appeared to have the same risk of non-melanoma mortality as age- and gender-matched controls taken from a midpoint survey of US census statistics. The complete follow-up of patients, the high rate of follow-up in patients, the ease of diagnosis of liver metastases, and the access to both death certificates and autopsy data suggest that erroneous cause of death assignment was not significant. Thus, uveal melanoma patients did not appear to have an altered life expectancy as a result of nonmalignant causes. The high risk of death from non-melanoma causes in this older age-group suggests that the cause of death should be accurately assessed in all survival analysis. Statistical analyses should account for competing causes of death. Treatment comparisons using melanoma-related mortalities should have a greater power than those that ignore the cause of death. Visual loss and ocular morbidity were significant problems in irradiated patients. In patients who had retained their eye 10 or more years after treatment, only 23% had visual acuities of 20/40 or better. In 18 patients with tumors less than 6 mm in thickness that were more than 3 mm from the optic nerve and fovea, 13 (72%) retained this level of vision. Tumor thickness and proximity to the optic nerve and fovea were the major risk factors for visual loss.³³ For each of these 2 variables, a change of 1 mm significantly increased the relative risk of halving the Snellen visual acuity, by 1.23 and 1.10, respectively. As shown in Fig 6, visual loss developed more frequently in the first few years after therapy.³⁴ The time course of early posttreatment visual loss is more consistent with an intrinsic tumor effect, since direct visually destructive radiation damage is usually delayed for 2 to 3 years after therapy.³⁵ In this and previous reports from our group, we have not noted a significant difference in visual loss or morbidity between 50 and 80GyE of helium radiation doses.²⁰

Ocular morbidity was due to both inherent tumor properties (location and thickness) and treatment parameters (anterior segment dose). As an example, in tumors greater than 10 mm in thickness, 13 of 28 (46%) eventually required enucleation. Retrospective analyses demonstrate that the percentage of the anterior segment in the entrance beam was a major predictive factor for the development of both neovascular glaucoma and cataract.¹⁸ The former complication has almost always developed within the first 3 years after treatment, while visually significant cataract continues to develop at a nearly constant rate of approximately 7% per year. A similar profile for cataract development was noted by Gragoudas and colleagues.³⁶ In a study of uveal melanoma patients treated with cobalt plaques,³⁷ 84% of eyes had cataracts develop after 10 years. In our current proton studies, given the above data, we have tried to avoid as much anterior segment as possible, often using multiple beams that overlap on the tumor and minimize entrance through the anterior segment. We predict,

on the basis of these data, that this newer approach may lower the incidence of neovascular glaucoma. Similarly, we would expect that minimization of the anterior segment in the treatment port should decrease the enucleation rate after particle radiation. In this series, almost three quarters of enucleations were due to neovascular glaucoma (NVG)-related pain; similar data have been reported by others.⁵ Several other complications have been reported after particle irradiation, but these are less troubling (localized eyelash loss) or quite uncommon, such as scleral necrosis in less than 1% of cases.

These long-term follow-up data on patients undergoing helium ion irradiation demonstrate that there is good local control and that good visual outcomes can be achieved, especially with tumors that are not too thick and away from the optic nerve and the fovea. Longer follow-up has not shown a significantly decreased local tumor control rate.

Unfortunately these long-term radiation data also point out several limitations to current radiation therapies. One, at all helium doses used, there was no difference in visual outcome or ocular morbidity. In an ongoing proton treatment trial, we have noted that a further reduction of radiation dose does not appear feasible, since there is an increased local failure rate. It is likely that some of the radiation complications we have reported, especially neovascular glaucoma, can be diminished by minimizing the amount of the anterior segment in the entrance beam. Unfortunately, for both tumor-related mortality and morbidity, tumor size and location are important parameters, and neither of these are affected by alterations in treatment.

Helium ion radiation of uveal melanoma is associated with good local tumor control and reasonable retention of treated eyes 10 years after therapy. In tumors that are less than 6 mm thick and greater than 3 mm from the optic nerve and fovea, many retain excellent vision. As seen in this study, the death rate of patients surviving uveal melanoma is similar to that of the general population. Approximately one half of the patients die of nonmelanoma-related causes.

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DISCUSSION

ALAN H. FRIEDMAN, MD. This paper reports on the experience and long-term results of patients treated with helium ion irradiation for uveal melanoma. There is much new and useful information that is noted in the paper. On the one hand, 22.4% of eyes were enucleated by 10 years; 95.4% of the eyes apparently had local control at 10 years; and almost half of the patients were dead after 10 years. The authors note that cataract formation is greater in proportion to a greater amount of radiation. Consequently, the greater amount of radiation delivered to the anterior segment, the higher the likelihood of neovascular glaucoma developing. The rate of visual loss increases as long as the patients are followed.

On the other hand, there are critical comments that must be addressed. First, the authors present their paper as a retrospective review of patients treated with helium ion therapy, yet they do not report on their selection bias and why these eyes were chosen to have helium ion and no other technique. The nonrandomized selection bias is an important point, and it is a defect in this paper as well as most papers published on uveal melanoma. How representative is this group? The answer to this question lies in the need for prospective, randomized trials to really compare techniques.

Second, the dose of radiation in the study varied significantly from 50 to 60 Gy to 70 to 80 Gy. The group is not homogenous.

Third, the authors state that all patients had "standardized tests," including medical investigations, at routine intervals. Is it possible that all 218 patients had every test done within a routine period? If not, how accurate is the data collection in general?

Fourth, the authors state that about half the patients' deaths were

malanoma-related, while the other half were not. Without autopsy confirmation, it is nearly impossible to know the true cause of death. This defect is common in most other melanoma papers, so that the reader does not really know the true cause of death.

Fifth, 84.9% of patients were symptomatic from melanoma. This is in contradistinction to most studies where melanomas were detected on routine examination and not based on symptoms. The question of selection bias is again raised.

The authors demonstrate that there is a relationship between local tumor control and patient survival. They do not reference Abramson and coworkers, who were the first to point this out.

The authors lump together *all* patients for survival, thus including patients with mean tumor thickness ranging from 1.3 to 14.7 mm and largest basal diameter ranging from 5 to 24 mm. They note that the larger the tumor, the more likely the patient's chance is of dying. By lumping all these disparate tumor sizes together, the authors have failed to assess the effectiveness of this technique for varying sizes of tumors.

The most important question for all ophthalmologists is, How does this technique compare with other brachytherapy for melanoma? Within the limitations of the study, it seems that the 50% survival rate is similar to that of brachytherapy; the 22.4% enucleation rate is higher than with brachytherapy; and the 90% tumor control is similar to that with brachytherapy. This suggests that helium ion therapy is as effective as radioactive plaque therapy and that the side effects of the 2 therapies are somewhat different. There is no evidence to conclude that helium ion therapy produces a higher rate of tumor control. Assuming that helium ion therapy and brachytherapy are similar, the lack of a prospective randomized study does not answer the important question of how the techniques compare with enucleation.

JACK COLEMAN, MD. I would like to compliment Dev on the duration of this study and the importance of long-term follow-ups. The questions I have relate to 2 aspects of tumor treatment. One has to do with the need for studies of ancillary treatments to either potentiate the effect of therapy or to reduce the level of radiation. We have used hyperthermia as an adjunctive treatment with radiation for over 12 years. Our results with hyperthermia plus radiation and hyperthermia plus enucleation were published this month in *Retina*.¹ The importance of long term follow-up is clear, as Devon's paper indicates. Our series is small for each arm but the follow-up is 7 to 10 years. The percentage of complications we have seems to be lower than the present paper but mostly these patients had cobalt plaque and radiation doses perhaps at the lower edge of the treatment range. Our analysis of our results and conclusions indicated a positive sur-

vival trend for combined hyperthermia and radiation. Our small series made it impossible to get a true statistical significance. We found that we reduced the relative risk by 32% by using the combined therapies. Regression models with acoustic tissue typing added, which gives us an idea of the type of melanoma risk, added to the positive trend. This added value of stratifying for risk with ultrasonic tissue analysis of features that appear to correspond to vascular loops and networks has previously been reported.^{2,3}

This brings up my second question. Not only is it the tumor type but it is also the volume of the tumor that correlates strongly with risk and more so than any univariate measurement.

My other question for you Dev, is that you mentioned in your talk that you thought tumor diameter was more important than volume. All of our data indicate that volume is a far more important measurement than using any single feature like height or diameter. In our data, we found a 16% higher survival when using combined therapies, and my questions really are, "Do you at this point in time still think that the single treatment modality is as effective as combined therapies, and secondly, how were your data derived that support a univariate measurement over volume as a risk model?"

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DENNIS M. ROBERTSON, MD. Devron, I congratulate you on providing some information about long-term follow-up of the helium treated patients. In your discussion, you made reference to the Collaborative Ocular Melanoma Study (COMS), indicating that the end point of the COMS was all-cause mortality. That indeed is true, but, in addition, the COMS, in a very elegant way, is collecting data that relates to tumor specific mortality. There are major differences between the COMS and your study: For example, unlike your study, the COMS is a randomized prospective study. Unlike your study, the COMS excluded all patients who had previously been diagnosed with another cancer (exclusive of basal cell carcinoma and carcinoma in situ of the cervix). The COMS found that approximately 15% of the patients who were evaluated with choroidal melanoma had some other form of cancer which excluded them from the

study. You report that about 1 out of 3 eyes develop neovascular glaucoma subsequent to radiation treatment. I am pleased you are directing attention toward minimizing some of these severe anterior segment complications seen following helium ion radiation.

RICHARD FORSTER, MD. I was struck by the young age of the patients in this particular study. Fifty-eight was the average age. How, in the long term, did the patients who were over the age of 58 or 60 compare to those who were younger? Did they have a different survival rate. I think this is an opportunity to compare the younger and the older age groups. Were the patients selected for this type of therapy because they were younger than the average age of a patient with ocular melanoma? Perhaps I am mistaken; is the national average for the age of patients with ocular melanoma 58?

DEVRON CHAR, MD. I want to thank all of the discussants for raising several good points. First, the mean age of our treated cohort is approximately 60 years old. Some reports have noted more older patients, and being elderly is a risk factor for poorer uveal melanoma outcome. Older patients had slightly larger tumors. Even with multivariate analysis, the relative importance of those separate parameters is uncertain.

Dr Robertson made a good point that the COMS will also assess other secondary endpoints in addition to all that cause of mortality. We believe that the significant incidence of non-melanoma deaths in this treatment cohort makes that choice of a primary end point a poor one.

Dr Coleman has raised a number of important issues. In our study, melanoma volume in a Cox's multivariate model was a less important prognostic factor than largest tumor diameter. I can not explain this difference between different data sets. I know Dr Coleman's work, and I respect it. We came up with a different result.

The use of adjunctive therapies in uveal melanoma is interesting, and we have published a number of animal studies in that area. My impression is different than Dr Coleman's. I suspect that the main impact of ancillary therapy will possibly be to decrease tumor-related morbidity. As Dr Coleman's group has published, they observed no significant difference in survival with brachytherapy alone as compared with brachytherapy and hyperthermia.

Dr Friedman has raised several points, some of which I can congenitally address and others I cannot. He was bothered by the fact that we stated that tumor control with charged particles was better without randomized study data. In fact, we published a dynamically balanced prospective study looking exactly on that issue several years ago. Charged particles statistically had a significantly better local control rate than plaques with a

p-value of <0.001.

Several authors prior to Abramson and colleagues noted the adverse systematic effect of local radiation failure.

By definition, to be managed in any regional referral center always entails some degree of selection bias. In terms of the patients we entered in this study, the entrance criteria were clearly delineated. We treat, with the intent to preserve the globes, over 80% of the eyes we see with non-enucleation techniques.

The radiation dose used for uveal melanoma therapy was probably not an issue in this study. Many years ago, prior to our prospective randomized trial, we performed a phase I/II dose searching study with 50, 60, 70 and 80 GyE of helium ions, and we found no difference in complication, control, or enucleation rates. When we attempted to lower the dose much further, we had some local failures. The radiation doses described in this paper are not an issue.

We were very conservative in our definition of uveal melanoma-associated mortality. We had death certificates on 92 of 102 patients who died of metastatic disease. Of 10 patients who did not have post-mortem examinations, we know 5 died of metastatic melanoma. While Dr Friedman thought this was important, I think this was not a major problem.

Finally, I believe Dr. Freedman is confused on statistical concepts. In truth, by including all tumors, regardless of size in the analysis, it improves our ability to assess the effects of radiation on both morbidity and tumor-related mortality.

I very much appreciate all the comments made and thank you for allowing me to present the data.